

CS 255 Business Requirements Document Template

System Components and Design

Purpose

- DriverPass is a driver training company that wants to help students better prepare for DMV driving exams.
- The purpose of this project is to create a web-based system that provides online training, practice tests, and driving lesson scheduling.
- The system must allow customers to purchase training packages, schedule driving lessons, complete online practice tests, and monitor their progress.
- The system must support multiple user roles, including customers, secretaries, administrators, instructors, and IT personnel.
- The system should provide secure access to customer information and training records.

System Background

- DriverPass identified that many students fail DMV driving exams because they rely only on studying previous tests.
- The company wants to provide a combination of online learning and behind-the-wheel instruction.
- Customers should be able to register for training packages and schedule driving lessons.
- The system should provide online practice tests and educational content.
- The system must maintain current DMV rules, policies, and test questions.
- The system should allow administrators to manage users, appointments, packages, and reports.
- The system requires a database, web application, reporting functionality, scheduling functionality, and DMV content integration.

Objectives and Goals

- Allow customers to register and manage accounts.
- Provide online classes and practice examinations.
- Allow customers to schedule, modify, and cancel driving lessons.

- Assign drivers and vehicles to customer appointments.
- Store customer information securely.
- Allow administrators to generate reports and audit user activity.
- Support password recovery functionality.
- Maintain up-to-date DMV training content.
- Allow managers to disable training packages when necessary.
- Provide access from desktop and mobile devices.

Requirements

Nonfunctional Requirements

Performance Requirements

- The system shall operate as a cloud-based web application.
- The system shall be accessible from any internet-connected computer or mobile device.
- The system shall support multiple concurrent users.
- The system shall load pages and retrieve information within a reasonable response time.
- The system shall maintain high availability during business operations.
- The system shall automatically update DMV content when new information becomes available.
- The system shall support report generation and export to Excel format.

Platform Constraints

- The application shall operate through standard web browsers.
- The application shall support desktop computers, laptops, tablets, and mobile devices.
- The system shall utilize a centralized database for storing customer, scheduling, and training information.
- The system shall require internet access for data updates and synchronization.
- The system shall operate within a cloud-hosted environment.
- The system shall support integration with external DMV information sources.

Accuracy and Precision

- The system shall distinguish between customers, instructors, secretaries, administrators, and IT administrators.
- The system shall use role-based permissions.
- The system shall accurately assign instructors and vehicles to appointments.
- The system shall track all reservation modifications and cancellations.
- The system shall generate audit reports showing user activity.
- The system shall notify administrators of suspicious activity.

Adaptability

- Administrators shall be able to add, modify, disable, and remove user accounts.
- Administrators shall be able to enable and disable training packages.
- The system shall support future feature expansion.
- The system shall support software updates with minimal downtime.
- The IT administrator shall have full access to user account management.
- The system shall adapt to DMV content updates.

Security

- The system shall require user authentication through usernames and passwords.
- The system shall use HTTPS encryption.
- The system shall encrypt sensitive customer and payment information.
- The system shall support password reset functionality.
- The system shall lock accounts after multiple failed login attempts.

- The system shall maintain audit logs.
- The system shall restrict access using role-based permissions.

Functional Requirements

- The system shall validate user credentials during login.
- The system shall allow customers to create accounts.
- The system shall allow customers to schedule driving lessons.
- The system shall allow customers to modify appointments.
- The system shall allow customers to cancel appointments.
- The system shall assign instructors and vehicles to appointments.
- The system shall provide online training content.
- The system shall provide online practice exams.
- The system shall track test scores and progress.
- The system shall allow secretaries to schedule appointments.
- The system shall generate management reports.
- The system shall allow administrators to manage users.
- The system shall allow administrators to disable training packages.
- The system shall allow IT administrators to reset passwords.
- The system shall update DMV content automatically.
- The system shall export reports to Excel.

User Interface

- The system shall provide a web-based interface.

- Customers shall access the system through desktop and mobile browsers.
- Customers shall be able to schedule appointments and take practice tests.
- Customers shall be able to view lesson schedules and training progress.
- Secretaries shall be able to manage customer records and appointments.
- Instructors shall be able to view schedules and enter lesson notes.
- Administrators shall be able to manage users and reports.
- IT administrators shall be able to manage accounts and permissions.
- The interface shall display test scores, completion status, and lesson history.

Assumptions

- Customers have internet access.
- Customers have valid payment methods.
- Users have basic computer skills.
- DMV information can be accessed electronically.
- Cloud hosting services are available.
- Drivers and vehicles are available for scheduling.

Limitations

- The system requires internet connectivity.
- Offline data modifications are not supported.
- The system depends on DMV data availability.
- Package customization beyond enabling and disabling is outside the current project scope.

- The system depends on accurate user-entered information.
- Third-party payment processors may introduce external dependencies.

Gantt Chart HTML

